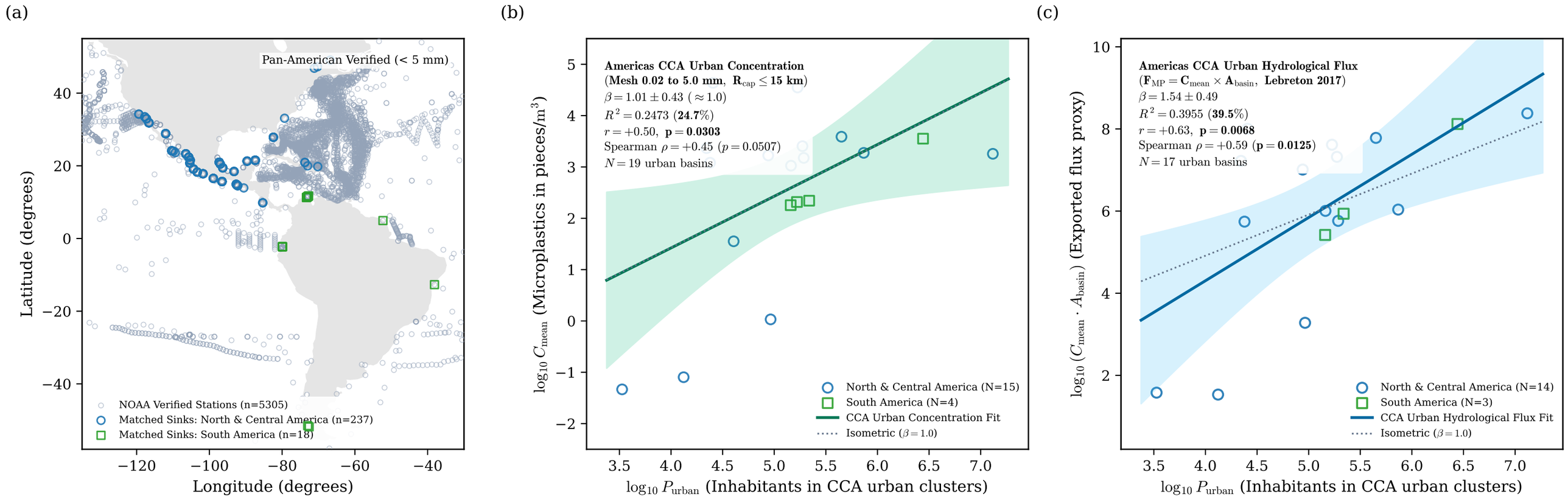

(a)
Pan-American Verified (< 5 mm)
NOAA Verified Stations (n=5305)
Matched Sinks: North & Central America (n=237)
Matched Sinks: South America (n=18)
Latitude (degrees)
Longitude (degrees)
40
20
0
−20
−40
−120
−100
−80
−60
−40
(b)
Americas CCA Urban Concentration
(Mesh 0.02 to 5.0 mm, $R_{cap} \leq 15$ km)
$\beta = 1.01 \pm 0.43$ (≈ 1.0)
$R^2 = 0.2473$ (24.7%)
$r = +0.50$, $p = 0.0303$
Spearman $\rho = +0.45$ ($p = 0.0507$)
$N = 19$ urban basins
North & Central America (N=15)
South America (N=4)
CCA Urban Concentration Fit
Isometric ($\beta = 1.0$)
$\log_{10} C_{mean}$ (Microplastics in pieces/m^3)
$\log_{10} P_{urban}$ (Inhabitants in CCA urban clusters)
5
4
3
2
1
0
−1
−2
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
(c)
Americas CCA Urban Hydrological Flux
($F_{MP} = C_{mean} \times A_{basin}$, Lebreton 2017)
$\beta = 1.54 \pm 0.49$
$R^2 = 0.3955$ (39.5%)
$r = +0.63$, $p = 0.0068$
Spearman $\rho = +0.59$ ($p = 0.0125$)
$N = 17$ urban basins
North & Central America (N=14)
South America (N=3)
CCA Urban Hydrological Flux Fit
Isometric ($\beta = 1.0$)
$\log_{10}(C_{mean} \cdot A_{basin})$ (Exported flux proxy)
$\log_{10} P_{urban}$ (Inhabitants in CCA urban clusters)
10
8
6
4
2
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0